

Knowledge Representation

Lecture 4:

Reasoning about Actions IV

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Certain and default effects

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For example, consider the following two shooting actions (performed by two different shooters).

1. *Shooting the gun (by Bill) **always** makes it unloaded and, if the gun was loaded before the action, makes Fred dead.*
2. *Shooting the gun (e.g., by Jim) **always** makes it unloaded and, if the gun was loaded before, it **typically** makes Fred dead.*

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Unless otherwise is known, we assume that typical effects occur.

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Default effects might be also viewed from the standpoint of the qualification problem: since it is in general impossible to enumerate all conditions under which the action is performed successfully and brings given results, we state that it *usually* has these results under specified (not necessary completely) conditions.

Basic assumptions

- Inertia law.
- Complete information about all actions and all fluents.
- Non-determinism.
- Side effects (ramifications) of actions are admitted.
- Action can be unexecutable in some states.
- Actions may have *certain* or *typical* (preferable) effects.

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To represent this class of dynamic systems we will use action languages of the class *AD*.

Action Language $\mathcal{AD}$ – Statements

A *signature* of $\mathcal{AD}$ is a pair $\Upsilon = (\mathcal{F}, \mathcal{Ac})$ defined as before.

The set of statements in $\mathcal{AD}$ is that of $\mathcal{AR}$ extended by the *typical effect statement* of the form:

A typically causes α if π

This statement says that the action A , whenever performed in a state satisfying π , *typically* leads to a state satisfying α . The effect α is *preferred* – in contrast to nondeterministic effects, where none result is preferred.

E.g. SHOOT *typically causes* $\neg alive$ *if* loaded.

Semantics of $\mathcal{AD}$

A **state** is a mapping $\sigma : \mathcal{F} \rightarrow \{0, 1\}$

As before, $\sigma \models \alpha$ means that α holds in σ (σ is a **model** of α).

Definition 4.1. A **structure** for $\mathcal{AD}$ is a system

$$S = (\Sigma, \sigma_0, ResAb, ResN)$$

where

- Σ is a set of states
- $\sigma_0 \in \Sigma$ is the initial state
- $ResAb, ResN : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ are transition functions. $ResAb$ and $ResN$ are respectively an **atypical transition** and a **typical transition** functions.

Semantics of $\mathcal{AD}$ (cont.)

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- Given an action domain D , a state σ is a **state of D** iff $\sigma \models \alpha$ for every constraint (**always** α) in D .
- Put $Res = ResAb \cup ResN$ and define $\Psi_S : \mathcal{Ac}^* \times \Sigma \rightarrow \Sigma$ as follows:
 - $\Psi_S(\varepsilon, \sigma) = \sigma$
 - if $\Psi_S((A_1, \dots, A_n), \sigma)$ is defined, then
 $\Psi_S((A_1, \dots, A_n), \sigma) \in Res(A_n, \Psi_S((A_1, \dots, A_{n-1}), \sigma))$.

Semantics of $\mathcal{AD}$ (cont.)

- A value statement α *after* $A_1, \dots, A_n$ is *true in S* iff for *every* mapping Ψ_S define as before,
 - $\Psi_S((A_1, \dots, A_n), \sigma_0)$ is defined
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Semantics of $\mathcal{AD}$ (cont.)

In order to define a model of an action domain D , let us define the following auxiliary mappings:

- $Res_0 : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ is defined wrt *effect statements*. Precisely, for any action $A \in \mathcal{Ac}$ and for any state $\sigma \in \Sigma$, $Res_0(A, \sigma)$ is the set of all states $\sigma' \in \Sigma$ satisfying the condition:

$$(A \text{ **causes** } \alpha \text{ **if** } \pi) \in D \wedge \sigma \models \pi \implies \sigma' \models \alpha.$$

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- $Res^-(A, \sigma) : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ is defined wrt minimization of changes, i.e. for any action $A \in \mathcal{Ac}$ and for any state $\sigma \in \Sigma$, $Res^-(A, \sigma)$ is the set of all states $\sigma' \in Res_0(A, \sigma)$, for which the sets $New(A, \sigma, \sigma')$ are minimal.

Semantics of $\mathcal{AD}$ (cont.)

- $Res_0^+ : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ is defined as follows: for any action $A \in \mathcal{Ac}$ and for any state $\sigma \in \Sigma$, $Res_0^+(A, \sigma)$ is the set of all states from Res_0 satisfying:

$$(A \text{ *typically causes* } \alpha \text{ *if* } \pi) \in D \wedge (\sigma \models \pi) \implies (\sigma' \models \alpha).$$

Res_0^+ contains all states where both **certain** and **typical** effects hold.

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Res_0^+ contains all states where both **certain** and **typical** effects hold.

- For any action $A \in \mathcal{Ac}$ and for any state $\sigma \in \Sigma$,

$$ResN(A, \sigma) = \{\sigma' \in Res_0^+(A, \sigma) : New(A, \sigma, \sigma') \text{ are minimal}\}$$

$$ResAb(A, \sigma) = Res^-(A, \sigma) \setminus ResN(A, \sigma).$$

$ResN$ is the set of all expected (wrt minimal changes) states satisfying both certain and typical effects, while $ResAb$ is the set of expected states, where only certain effects hold.

Semantics of $\mathcal{AD}$ – Model

Definition 4.2. Let D be an action domain of $\mathcal{AD}$. A structure $S = (\Sigma, \sigma_0, ResAb, ResN)$ is a **model of** D iff

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- all value and observation statements in D are true in S
- for every action $A \in \mathcal{Ac}$ and for every state $\sigma \in \Sigma$, $ResAb$ and $ResN$ are defined as before.

Example 4.1: YSP

initially $alive \wedge \neg loaded$;

LOAD *causes* $loaded$;

SHOOT *causes* $\neg loaded$;

SHOOT *typically causes* $\neg alive$ *if* $loaded$.

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$$\begin{array}{ll} \sigma_0 = \{alive, \neg loaded\} & \sigma_2 = \{\neg alive, \neg loaded\} \\ \sigma_1 = \{alive, loaded\} & \sigma_3 = \{\neg alive, loaded\}. \end{array}$$

$$Res_0(\text{SHOOT}, \sigma_0) = \{\sigma_0, \sigma_2\}$$

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It is worth noting that whenever an action has certain effects only, they are viewed as **typical**.

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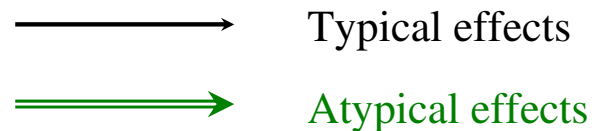
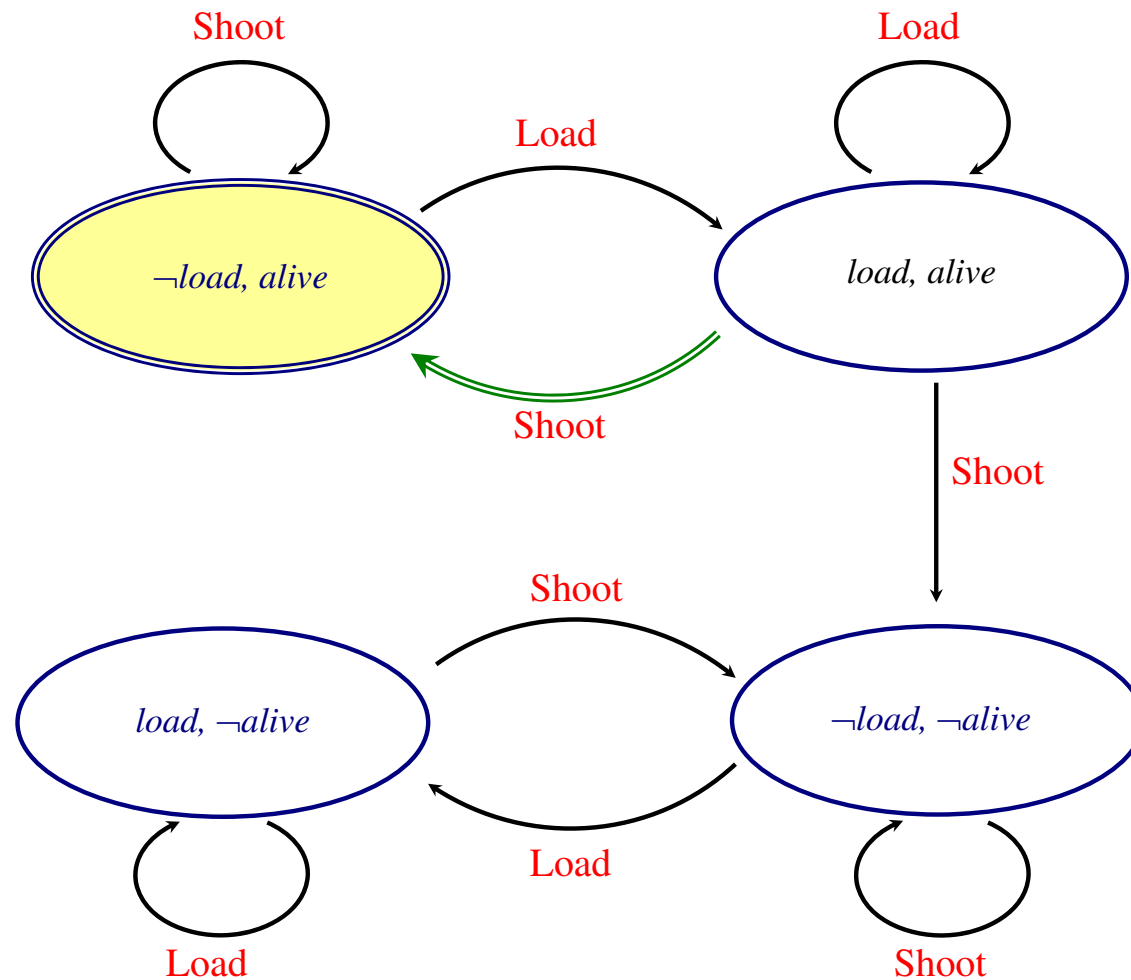
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Queries of $\mathcal{AD}$

Consider the following two queries:

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The 1st query states that α **always** holds after performing the sequence $(A_1, \dots, A_n)$ of actions (regardless of the way the actions were executed, i.e. typical or atypical).

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The 1st query states that α **always** holds after performing the sequence $(A_1, \dots, A_n)$ of actions (regardless of the way the actions were executed, i.e. typical or atypical).

The 2nd query says that α **typically** holds after performing the sequence $(A_1, \dots, A_n)$ of actions. Intuitively, “typically” is understood as *minimal atypical transitions* on paths determined by $(A_1, \dots, A_n)$.

Abnormality measure

Let $S = (\Sigma, \sigma_0, ResAb, ResN)$ be a structure for $\mathcal{AD}$.

For any action $A \in \mathcal{Ac}$ and for any two states $\sigma, \sigma' \in \Sigma$, define

$$\kappa_S(A, \sigma, \sigma') = \begin{cases} 0 & \text{iff } \sigma' \in ResN(A, \sigma) \\ 1 & \text{iff } \sigma' \in ResAb(A, \sigma) \\ +\infty & \text{otherwise.} \end{cases}$$

Abnormality measure (cont.)

For any mapping Ψ_S and for any sequence $(A_1, \dots, A_n)$ of actions

- $K_\Psi(A_1, \dots, A_n) = +\infty$ iff $\Psi_S((A_1, \dots, A_n), \sigma_0)$ is undefined

- $K_\Psi(A_1, \dots, A_n) = \sum_{i=1}^n \kappa_S(A_i, \sigma_{i-1}, \sigma_i)$, where $\sigma_i = \Psi_S(A_i, \sigma_{i-1})$

Abnormality measure (cont.)

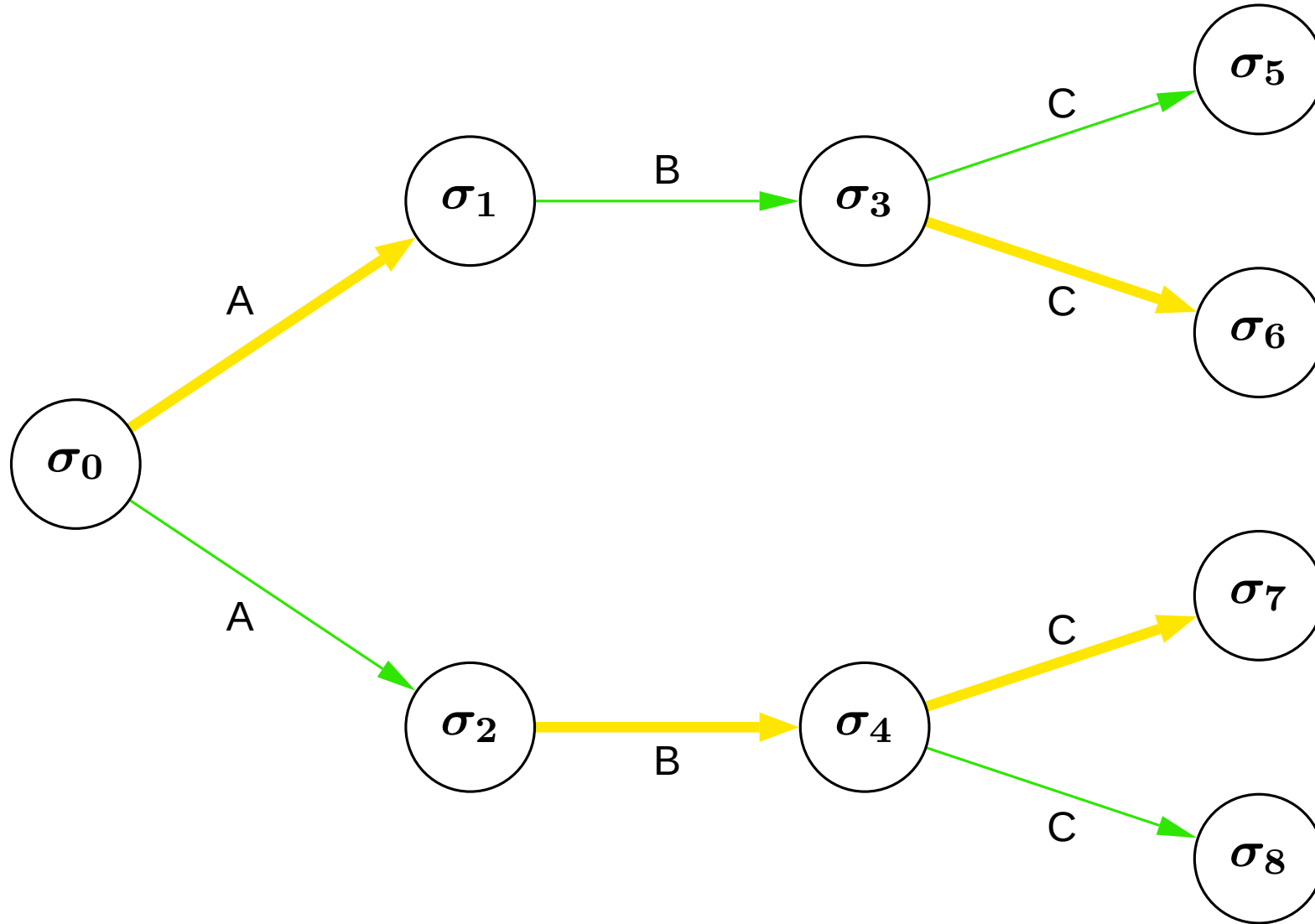
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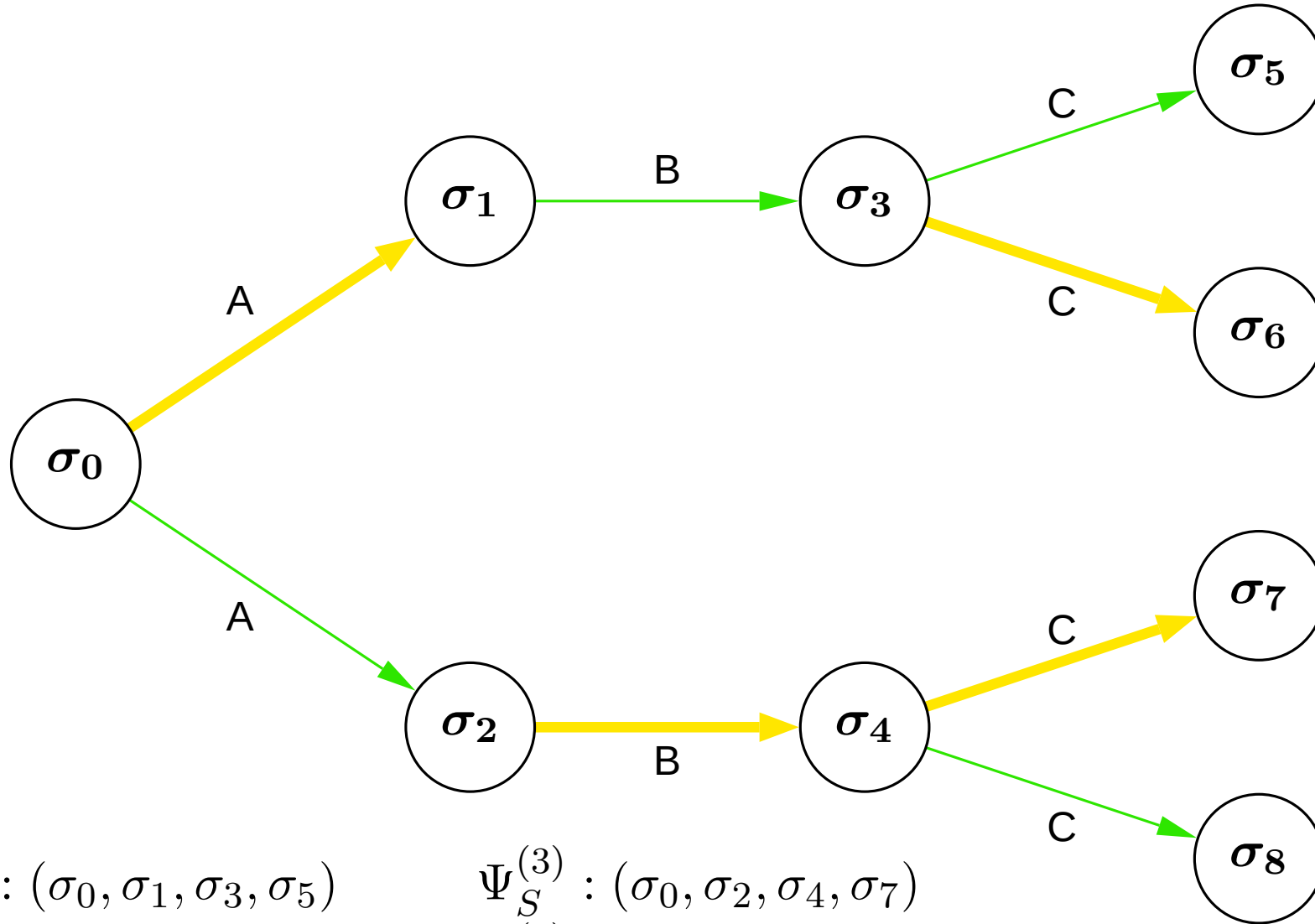
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For any sequence $(A_1, \dots, A_n)$ of actions, $n \geq 1$, let $\Pi_S(A_1, \dots, A_n)$ stand for the set of all Ψ_S mappings for which $K_\Psi(A_1, \dots, A_n)$ are minimal.

Mappings Ψ_S and sets $\Pi_S(A_1, \dots, A_n)$



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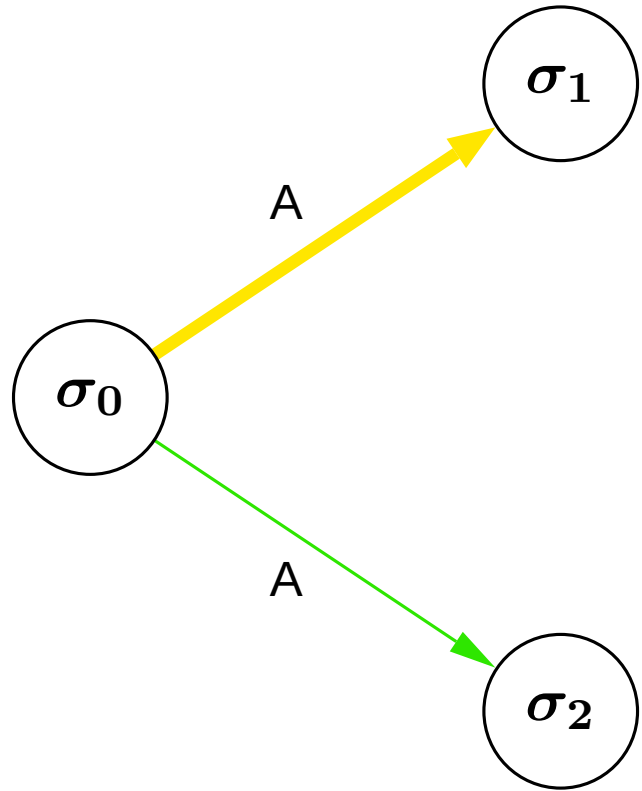
$$\Psi_S^{(1)} : (\sigma_0, \sigma_1, \sigma_3, \sigma_5)$$

$$\Psi_S^{(2)} : (\sigma_0, \sigma_1, \sigma_3, \sigma_6)$$

$$\Psi_S^{(3)} : (\sigma_0, \sigma_2, \sigma_4, \sigma_7)$$

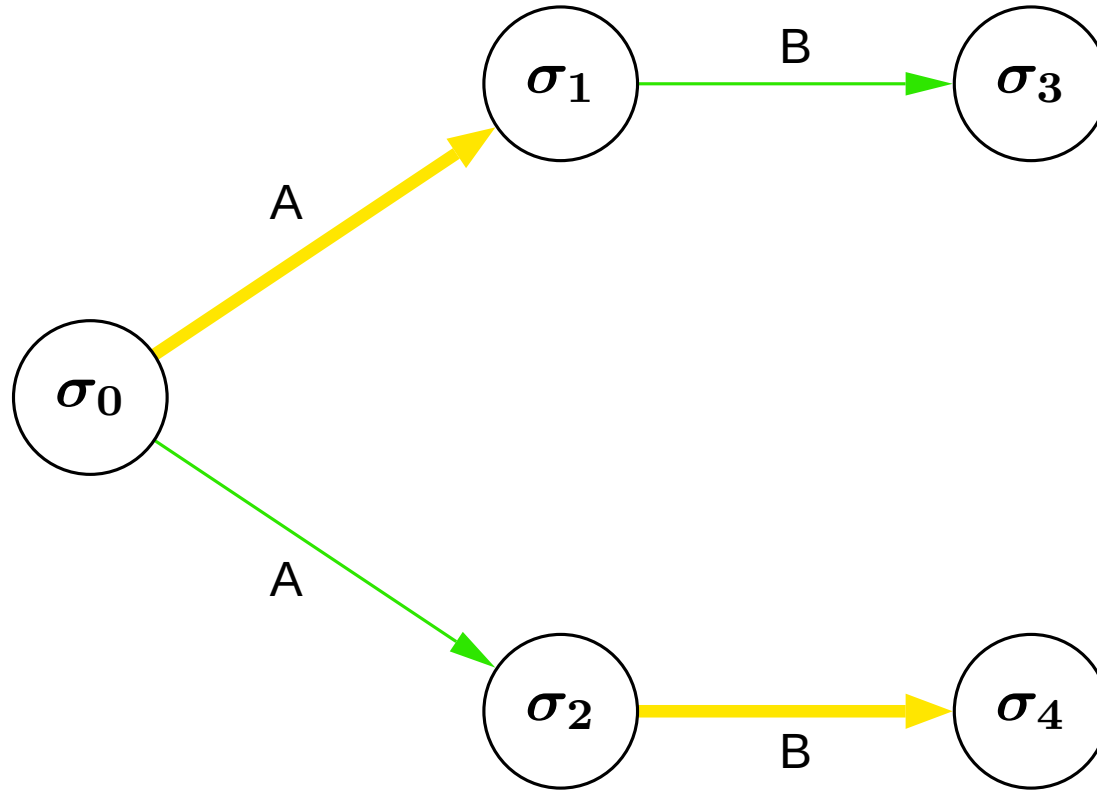
$$\Psi_S^{(4)} : (\sigma_0, \sigma_2, \sigma_4, \sigma_8).$$

Mappings Ψ_S and sets $\Pi_S(A_1, \dots, A_n)$



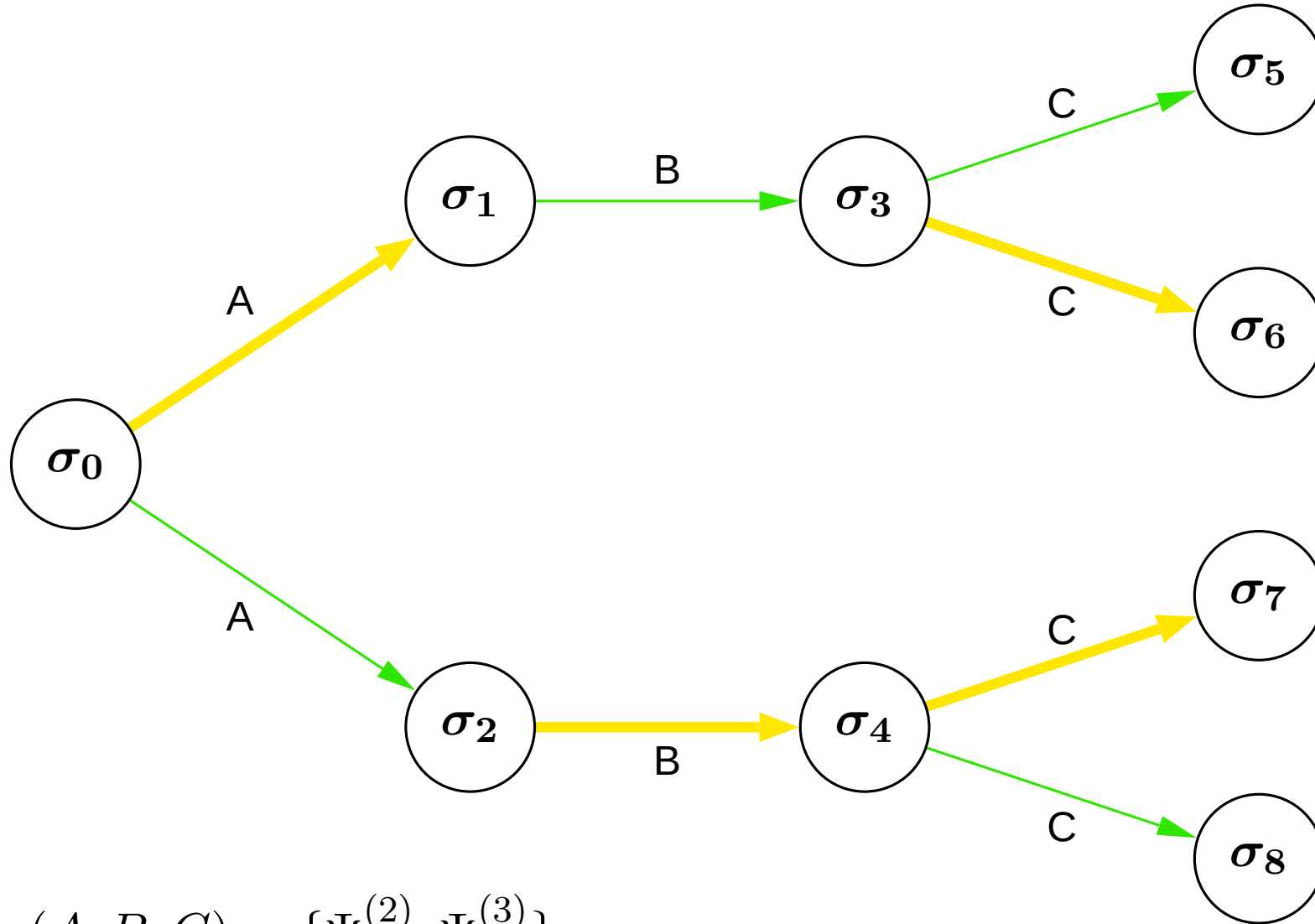
$$\Pi_S(A) = \{\Psi_S^{(1)}, \Psi_S^{(2)}\}.$$

Mappings Ψ_S and sets $\Pi_S(A_1, \dots, A_n)$



$$\Pi_S(A, B) = \{\Psi_S^{(1)}, \Psi_S^{(2)}, \Psi_S^{(3)}, \Psi_S^{(4)}\}.$$

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$$\Pi_S(A, B, C) = \{\Psi_S^{(2)}, \Psi_S^{(3)}\}.$$

Consequences of D

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- if Q is of the form A **typically causes** α **after** $A_1, \dots, A_n$, then $D \models Q$ iff $\Psi_S^*((A_1, \dots, A_n), \sigma_0) \models \alpha$ for every model $S = (\Sigma, \sigma_0, ResAb, ResN)$ of D and for every $\Psi_S^* \in \Pi_S(A_1, \dots, A_n)$.

Example 4.1: YSP (cont.)

initially $alive \wedge \neg loaded$;

LOAD *causes* $loaded$;

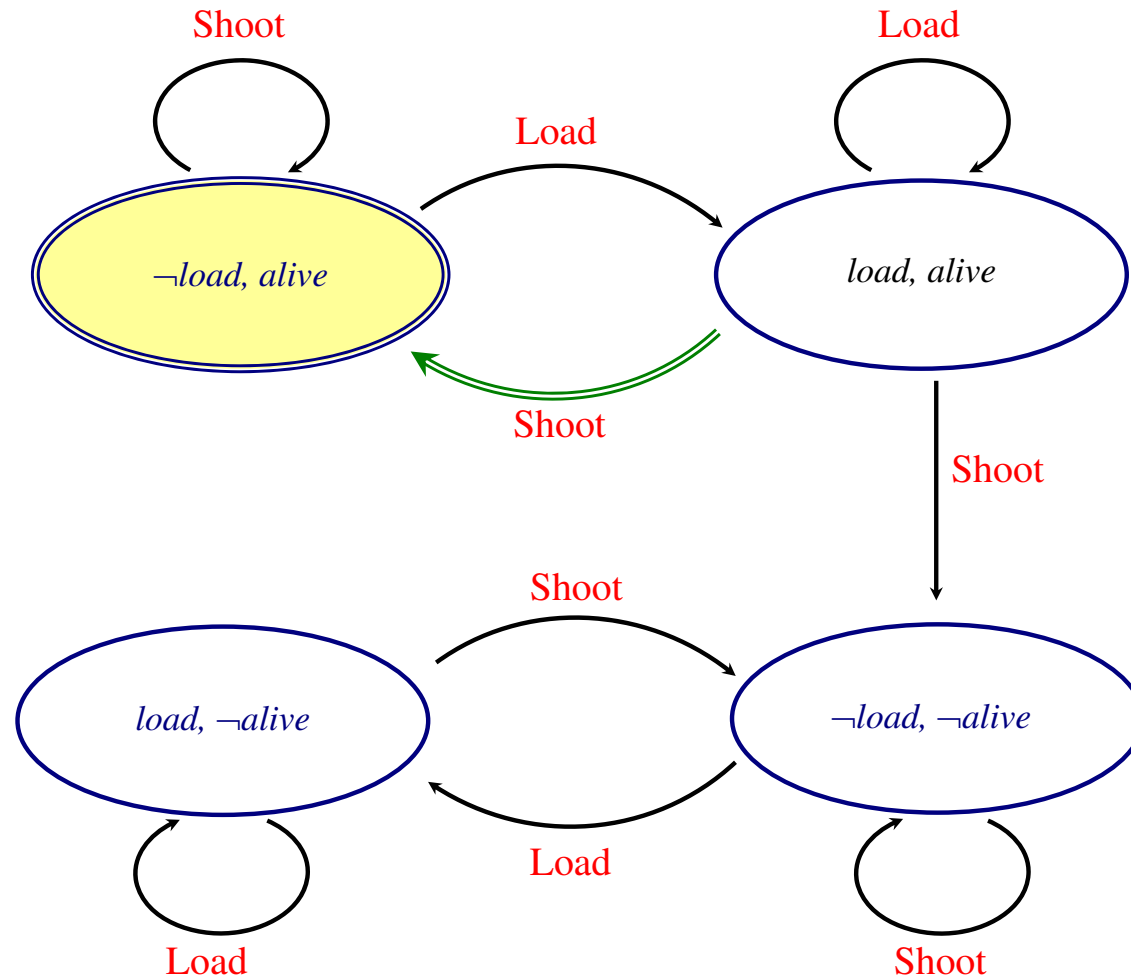
SHOOT *causes* $\neg loaded$;

SHOOT *typically causes* $\neg alive$ *if* $loaded$.

Note that

$D \models \textit{typically } \neg alive \textit{ after } LOAD, Shoot$.

Example 4.1: YSP (cont.)



Final remark

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- Determine models of D^- in the sense of $\mathcal{AD}$. Let $S^- = (\Sigma^-, \sigma_0, ResN^-, ResAb^-)$ be such a model, where σ_0 satisfies all constraints. Put:
 - $\Sigma \subseteq \Sigma^-$ is the set of all states, where all constraints hold.
 - For any $A \in \mathcal{Ac}$ and for any $\sigma \in \Sigma$,
$$ResN(A, \sigma) = ResN^-(A, \sigma) \cap \Sigma \text{ and}$$
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Thank you for your attention!

Any questions are welcome.